

**PERSONNEL QUALIFICATIONS
FOR**

**DEVELOPMENT, INTEGRATION AND MAINTENANCE OF SPECIAL RESEARCH AND
DEVELOPMENT EQUIPMENT FOR THE POMONKEY FACILITY**

Key Personnel:

RADIO FREQUENCY & MICROWAVE SYSTEMS TECHNICIAN
ELECTROMECHANICAL SPECIALIST
MODELMAKER

Other Personnel:

ANTENNA SUPPORT PERSONNEL

DRAFT

RADIO FREQUENCY & MICROWAVE SYSTEMS TECHNICIAN (Key Personnel)

EDUCATION:

Certification from a four-year technical school and a High School Diploma.

EXPERIENCE:

Minimum of twenty-five years of experience as a Radio Frequency and Microwave Systems Technician. Must also include twenty years of experience climbing towers or similar structures and using OSHA approved safety equipment at high elevations (up to 135 feet).

KNOWLEDGE AND SKILLS:

The Radio Frequency & Microwave Systems Technician is required to have hands-on experience and a practical knowledge of radio frequency (3 – 300 MHz), microwave (300 – 40,000 MHz), electrical, and mechanical theory and principles as applied to full motion parabolic (FMP) antenna systems and scientific radio frequency and microwave equipment, including the maintenance and repair of such antenna systems and equipment. Work demands ability to perform antenna system, radio frequency component, and microwave component test, measurement, and calibration. The Technician must be able to use a wide variety of electronic test equipment as well as mechanical measurement equipment to verify proper operation and test modifications of, or improvements to, the Facility's FMP antenna systems. In addition, knowledge in troubleshooting and repairing electronic and servo mechanisms to the component level is required to locate sources of defective operation or cause of breakdown of these antenna systems or related laboratory equipment.

This person must be proficient in the use of the following equipment:

- Spectrum Analyzers
- Microwave Power Meters
- Radio Frequency and Microwave Function Generators
- Line Elements
- Time Domain Reflectometers
- Digital Volt Meters
- Servo Response Recorders
- Soldering/Desoldering Stations

REQUIREMENTS:

The Radio Frequency & Microwave Systems Technician shall install, repair, adjust, specify, test and maintain scientific equipment designed for radio frequency and microwave systems such as: antenna feeds, low noise amplifiers, power amplifiers, high power amplifiers (both solid state and travelling wave

tube), radio frequency and microwave receivers, passive microwave components (i.e. waveguide, coaxial transmission line, couplers, power splitters), antenna servo control systems, and remote sensing systems. The Technician shall perform system-level troubleshooting to determine point of failure of microwave components and/or circuits. The person must also be capable of specifying and performing precision microwave transmission line assembly, installation, and testing of both waveguide and coax transmission lines. For these transmission lines the Technician must be able to confirm that the transmission line assemblies meet specification and if not, take appropriate corrective action.

The Technician shall also install, troubleshoot, adjust, and repair electronic systems related to the movement and tracking capabilities of full motion, large parabolic tracking antennas. This includes servo electronics, drive motors and their controllers, and antenna control units.

In addition, the Radio Frequency & Microwave Systems Technician should be thoroughly experienced in the following areas:

- Experienced in the maintenance and operation of the electronic portion of servo drive systems utilized in elevation over azimuth tracking antenna pedestals.
- Specifically experienced in the electronic operation and maintenance of very large full motion parabolic antenna systems such as Vertex Communication Systems Inc. Antenna Tracking Systems.
- Specifically experienced in the electronic operation and maintenance of Scientific-Atlanta Inc. full motion elevation over azimuth Antenna Tracking Systems.
- Experienced in the electronic operation and maintenance of Kingpost mount Elevation over Azimuth antenna systems.
- Experienced in the operation and troubleshooting of both solid state and travelling wave tube high power amplifiers manufactured by Empower, ETM, MCL, and CPI.
- Experienced with the assembly and installation of precision waveguide and coaxial cable assemblies as applied to JPL's Deep Space Network requirements.
- Experienced with waste oil management procedures per NRLINST 6240.7B, *NRL Oil/Used Oil Management* and Title 40 Code of Federal Regulations (CFR) Part 112.7 and 40 CFR 112.8, *Oil Pollution Prevention*.

DUTIES:

As indicated in the statement of work.

ELECTROMECHANICAL SPECIALIST (Key Personnel)

EDUCATION:

Certification from a four-year technical school and a High School diploma/GED. Possession of a Commercial Driver's License is also desired.

EXPERIENCE:

Minimum of ten years of experience as an electromechanical specialist including ten years of experience climbing towers or similar structures and using OSHA approved safety equipment at high elevations (up to 135 feet).

KNOWLEDGE AND SKILLS:

The Electromechanical Specialist is required to have a broad knowledge of mechanical and electrical principles as applied to a variety of systems including full motion, large aperture parabolic antenna systems, prime focus antenna focal point mechanical systems, physical access systems, closed circuit camera systems, IP based telephone systems, and scientific test and measurement equipment required to test, calibrate, and troubleshoot the above mentioned systems.

Work demands ability to use a wide variety of electronic test equipment as well as mechanical measurement equipment to verify proper operation, troubleshoot, test modifications of, or improvements to these various types of systems. In addition, knowledge in troubleshooting the electromechanical portion of servo mechanisms to the component level is required to locate sources of defective operation or cause of breakdown of machinery or laboratory equipment and to use special skills to repair and maintain this equipment in the best working condition.

This person must be proficient in the use of the following equipment:

- Spectrum Analyzers
- Power Meters
- Function Generators
- Line Elements
- Time Domain Reflectometers
- Digital Volt Meters
- Servo Response Recorders
- Soldering/Desoldering Stations

REQUIREMENTS:

The Electromechanical Specialist shall install, troubleshoot, repair, adjust, test and maintain scientific equipment designed on electromechanical principles such as pedestal and dish structures of full motion, large aperture antennas and their mechanical drive system components, microwave components related to these antennas (i.e. transmitters, waveguide, transmission lines), servo control systems, remote sensing systems and thermal systems, as well as, hydraulic, pneumatic, gear driven and thermal systems. Using the knowledge gained through work experience, or OJT, of electronics, as well as, mechanical and hydraulic principles, locates sources of defective operation or causes of system errors. This person must be capable of disassembly of equipment, repairs or manufactures and replaces defective machine parts, adjust and re-align gear trains, shafts, bearings, and similar parts, adjust clearances and resets any maladjustment on any mechanical device utilized in servo control systems. This person must also be capable of precision microwave transmission line assembly and installation, as well as basic troubleshooting of existing microwave circuits. Use the standards set forth in the NASA Deep Space Network for the installation of all waveguide components. In addition, the Electromechanical Specialist should be thoroughly experienced in the following areas:

- Experienced in the electromechanical maintenance and operation of servo drive systems utilized in elevation over azimuth tracking antenna pedestals.
- Demonstrated experience in the design, fabrication, and installation of devices utilizing electromechanical components for control.
- Specifically experienced in the electromechanical operation and maintenance of full motion large aperture parabolic antenna systems such as Vertex Communication Systems Inc. Antenna Tracking Systems.
- Experienced in the electromechanical operation and maintenance of Scientific-Atlanta Inc. full motion elevation over azimuth Antenna Tracking Systems.
- Experienced in the electromechanical operation and maintenance of Kingpost mount Elevation over Azimuth antenna systems.
- Experienced with the assembly and installation of precision waveguide and coaxial cable assemblies as applied to JPL's Deep Space Network requirements.
- Experienced with the installation and maintenance of various size antenna radomes.
- Experienced with Backup Generator Systems including installation, test and repair.
- Experienced with electromechanical systems related to vehicle access gates including installation, test and repair.
- Specifically experienced with Avaya IP based telephone systems including installation, test and repair.
- Experienced with installation and troubleshooting of AC and DC power circuits.
- Experienced with installation and troubleshooting of IP based closed circuit camera systems.

- Experienced with waste oil management procedures per NRLINST 6240.7B, *NRL Oil/Used Oil Management* and Title 40 Code of Federal Regulations (CFR) Part 112.7 and 40 CFR 112.8, *Oil Pollution Prevention*.

DUTIES:

As indicated in the statement of work.

DRAFT

MODELMAKER (Key Personnel)

EDUCATION:

Full certification as a journeyman machinist and a High School diploma/GED. Possession of a Commercial Driver's License is also desired.

EXPERIENCE:

Minimum of ten years of experience as a mechanical Modelmaker, designer, and fabricator using metal and wood, plus an additional six years as a machinist apprentice.

KNOWLEDGE AND SKILLS:

The Modelmaker is required to have a broad knowledge of mechanical, electromechanical and hydraulic principles as it concerns machinery and laboratory apparatus, maintenance and repair. Work demands ability to use the full variety of the regular and specially adapted tools of the machinist trade plus the ability to locate the source of defective operation or cause of breakdown of machinery or laboratory equipment and to use special skills to repair and maintain this equipment in the best working condition.

This person must be proficient in the use of the following equipment:

- Engine Lathe
- Turret Lathe
- Milling Machine
- Drill Press
- Planer
- Shaper
- Grinder
- Boring Mill, Horizontal and Vertical

REQUIREMENTS:

The Modelmaker shall install, repair, adjust, test and maintain scientific equipment designed basically on mechanical principles. Using the knowledge gained through work experience, or OJT, of machines, mechanical and hydraulic principles, locates sources of defective operation or causes of system errors. The Modelmaker must be capable of disassembly of equipment, repairs or manufactures and replaces defective machine parts. Overhauls various interrelated systems in complex machinery such as power drive, transmissions, speed reduction or acceleration mechanisms. Repairs and dynamically balances shafts and other rotating parts. In addition, the Modelmaker should be thoroughly experienced in the following areas:

- Specifically experienced in the operation of a Hardinge EZ Vision, three-axis CNC mill.
- Experienced in the fabrication of fixtures for mounting feed horns and in the installation and precision alignment of feed horns for maximum efficiency as required by the statement of work.
- Experienced and proficient in the areas of precision machining, wood working, cable installation, and general site work as required by the statement of work.
- Experienced with waste oil management procedures per NRLINST 6240.7B, *NRL Oil/Used Oil Management* and Title 40 Code of Federal Regulations (CFR) Part 112.7 and 40 CFR 112.8, *Oil Pollution Prevention*.
- Experienced with the maintenance of various size antenna radomes.
- Experienced in the maintenance and repair of chain link fencing.

DUTIES:

As indicated in the statement of work, the Modelmaker must be capable of working to a tolerance of 0.0005 inches.

ANTENNA SUPPORT PERSONNEL

EDUCATION:

High School diploma/GED

EXPERIENCE:

Minimum of five years of experience in the large aperture parabolic dish antenna construction trade. This experience should include structural steel fabrication and welding. In addition, two years of experience climbing towers or similar structures using OSHA approved safety equipment at high elevations (up to 135 feet).

KNOWLEDGE AND SKILLS:

Antenna Support Personnel are required to have a broad knowledge of mechanical and electrical construction principles as it applies to antenna tracking systems. A background in mechanical assembly and steel construction is advisable. Work demands knowledge of basic troubleshooting of mechanical and electrical systems to locate potentially defective components.

This person must be proficient in the use of the following equipment:

- Hand Tools
- Power equipment
- Welders, MIG and TIG
- Digital Volt Meters
- Soldering/Desoldering Stations

REQUIREMENTS:

Antenna Support Personnel shall repair, adjust, test and maintain various antenna systems and support facilities.

This person must be capable of disassembly of equipment, repairs or manufactures and replaces defective machine parts, assist in the adjustment and re-aligning of gear trains, shafts, bearings, and similar parts.

In addition, Antenna Support Personnel should be experienced in the following areas:

- Experienced in the general operation and maintenance of Vertex Communication Systems Inc. Antenna Tracking Systems.
- Experienced in the general operation and maintenance of Scientific-Atlanta Inc. full motion elevation over azimuth Antenna Tracking Systems.
- Experienced in the general operation and maintenance of Kingpost mount Elevation over Azimuth antenna systems.

- Experienced with the installation and maintenance of various size antenna radomes.
- Experienced in the maintenance and repair of chain link fencing.
- Experienced with Backup Generator Systems operation.
- Experienced with waste oil management procedures per NRLINST 6240.7B, *NRL Oil/Used Oil Management* and Title 40 Code of Federal Regulations (CFR) Part 112.7 and 40 CFR 112.8, *Oil Pollution Prevention*.

DUTIES:

As indicated in the statement of work.

DRAFT